

PIPELINE FAILURE ROOT CAUSE ANALYSIS

Automated AI Incident Analysis & Resolution Report

Report ID:	RCA-0001	Created At:	2026-06-07 06:06:39 UTC
Severity:	CRITICAL	Confidence Score:	95%

1. Incident Summary

Pipeline execution failed with errors.

2. Root Cause Identification

The pipeline failed due to a `KeyError` exception: `'timestamp'` in `pipeline/handler.py` at line 15. This was directly introduced in commit `8c7a2b9f3d1e4e5f6a7b8c9d0e1f2a3b4c5d6e7f` ('Refactor dictionary response keys in ETL payload formatting') by `dev-engineer-lead`. The commit changed the payload mapping keys to use `'ts'` instead of `'timestamp'`, but downstream database loader script still expects the `'timestamp'` key, leading to a crash when it tries to extract `'timestamp'` from the payload dictionary.

3. Correlated GitHub Code Changes

Commit ID:
Changed Files: None

No obvious matching code changes found correlating with the error.

4. Recommended Resolution Path

Update the downstream database loader code in `etl_job.py` to expect `'ts'` instead of `'timestamp'`, or revert the dictionary key change in `pipeline/handler.py` to maintain backward compatibility.

5. Recommended Rerun/Retry Steps

1. Revert commit 8c7a2b9f or apply hotfix to pipeline/handler.py.\n2. Run local tests (pytest tests/test_handler.py) to confirm formatting structure.\n3. Push fix to main repository.\n4. Trigger pipeline rerun via standard workflow.